

Architecture 3-tiers

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Oct  1 11:34:00 CEST 2025 on tty1
root@VMBDD:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN g
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
    link/ether 08:00:27:e9:f1:3c brd ff:ff:ff:ff:ff:ff
    inet 172.17.21.20/16 brd 172.17.255.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fee9:f13c/64 scope link
        valid_lft forever preferred_lft forever
root@VMBDD:~# _
```

```
the exact distribution terms for each program
individual files in /usr/share/doc/*/copyright

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Oct  1 11:37:32 CEST 2025 on tty1
root@VMWEB:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN g
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
    link/ether 08:00:27:70:a3:6d brd ff:ff:ff:ff:ff:ff
    inet 172.17.21.10/16 brd 172.17.255.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe70:a36d/64 scope link
        valid_lft forever preferred_lft forever
root@VMWEB:~#
```

```
GNU nano 7.2 /etc/mysql/mariadb.conf.d/50-server.cnf
#
# These groups are read by MariaDB server.
# Use it for options that only the server (but not clients) should see

# this is read by the standalone daemon and embedded servers
[server]

# this is only for the mysqld standalone daemon
[mysqld]

#
# * Basic Settings
#

#user                 = mysql
pid-file              = /run/mysqld/mysqld.pid
basedir              = /usr
#datadir              = /var/lib/mysql
#tmpdir               = /tmp

# Broken reverse DNS slows down connections considerably and name resolve is
# safe to skip if there are no "host by domain name" access grants
#skip-name-resolve

# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address          = 0.0.0.0
```

Étape 1 : Schéma de l'architecture 3-tiers

- Tu dois représenter les **3 niveaux** :
 1. **Client (navigateur)** → accède à l'IHM via HTTP.
 2. **Serveur web** (Apache2 + PHP) → IP 172.17.21.10
 3. **Serveur BDD** (MariaDB) → IP 172.17.21.20

Étape 2 : Configuration MariaDB (VM bdd)

Édite la config MariaDB pour l'accès distant :

```
nano /etc/mysql/mariadb.conf.d/50-server.cnf  
→ Modifier :
```

```
bind-address = 0.0.0.0
```

Redémarre le service :

```
systemctl restart mariadb
```

Étape 3 : Création de l'utilisateur distant

Toujours sur la VM **bdd** :

```
GRANT ALL PRIVILEGES ON t_films.* TO 'titi'@'%' IDENTIFIED BY 'password123';  
FLUSH PRIVILEGES;
```

Étape 4 : Installation Apache + PHP (VM web)

Sur la VM **web** :

```
apt install apache2 php libapache2-mod-php php-mysql -y
```

Étape 5 : Test de connexion depuis la VM web

Installe le client MariaDB :

```
apt install mariadb-client -y
```

Teste la connexion :

```
mysql -u titi -h 172.17.21.20 -p
```

Étape 6 : Installation de phpMyAdmin

Toujours sur la **VM web** :

```
apt install phpmyadmin -y
```

- Choisir apache2 comme serveur web.
- Entrer le mot de passe root de MariaDB si demandé.

Étape 7 : Configurer phpMyAdmin pour se connecter à la VM bdd

Édite :

```
nano /etc/phpmyadmin/config.inc.php
```

Changer :

```
$cfg['Servers'][$i]['host'] = '172.17.21.20';
```

Étape 8 : Accès à phpMyAdmin

Depuis un navigateur :

```
http://172.17.21.10/phpmyadmin
```

Identifiants :

- **Utilisateur** : titi
- **Mot de passe** : password123

Étape 9 : Test d'une page PHP

Créer le fichier :

```
nano /var/www/html/db_test.php
```

Code :

```
<?php
$servername = "172.17.21.10";
$dbname = "t_films";
$username = "titi";
$password = "password123";

$conn = mysqli_connect($servername, $username, $password, $dbname);

if (!$conn) {
    die("Echec de connexion : " . mysqli_connect_error());
}
echo "Connexion réussie !";
mysqli_close($conn);
?>
```

Accéder dans un navigateur :

```
http://172.17.21.10/db_test.php
```